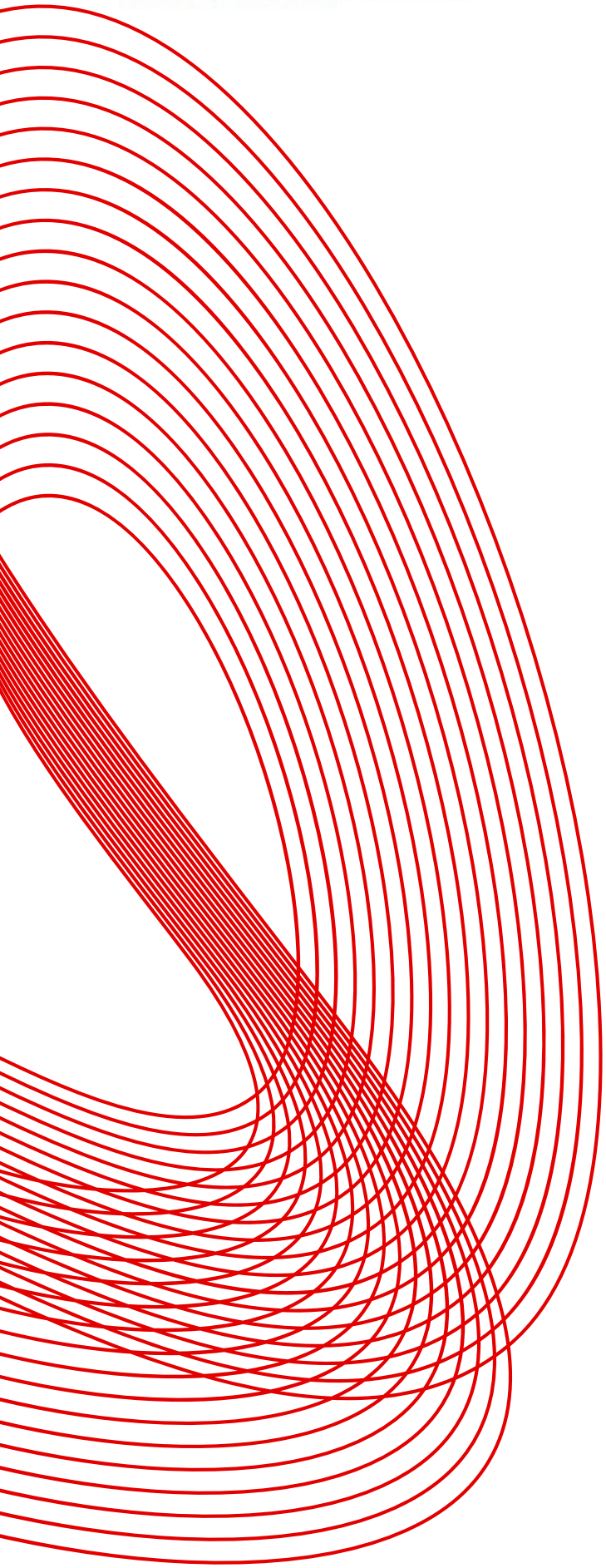
INSA INSTITUT NATIONAL DES SCIENCES APPLIQUÉES STRASBOURG

S9



# S-IoT Project Report :

## Drone swarm


**Written by :**

Emilien PEYON

Noé RAMDANI

**Class of 2022-2027**

# Contents

**Introduction** 5

**I Hardware** 6
1 Drone specifications . . . 6
2 Drone mechanical components . . . 6
A Structure . . . 6
B Propulsion system . . . 6
3 Drone electrical system . . . 7
A Flight controller . . . 7
B ESC (Electronic Speed Controllers) . . . 7
C Power supply . . . 7

**II Specifications** 8
1 Objective . . . 8
2 Functions . . . 8
3 Deliverables . . . 8

**III Bibliographical Studies** 9
1 UWB . . . 9
A Definitions . . . 9
B Location algorithm . . . 9
C Timing algorithm . . . 11
2 Measure processing . . . 13
A Extended Kalman Filter for relative localisation . . . 13
B UWB localization : benefits and limits . . . 15
C Data fusion . . . 15
3 Communication protocol . . . 17
A Xbee . . . 17
B UWB . . . 17
C ESP-NOW . . . 18

**IV Design and Implementation** 19
1 UWB . . . 19
A Algorithm choice . . . 19
B Calibration . . . 19
C Test . . . 20
D Filter . . . 23
2 Communication . . . 25
A Choice . . . 25
B ESP-NOW implementation . . . 25
3 Measure processing and formation control . . . 27
4 Visualization . . . 28
A Matlab . . . 28
B Architecture . . . 28
5 Simulation . . . 29
A ArduPilot and Mission Planer . . . 29
B Architecture . . . 30
6 Test . . . 31
A Simulation . . . 31

B Outdoor . . . . . 33

**Conclusion** **34**

**Calculations** **35**

**References** **37**

# List of Figures

III.1 TWR explanation diagram . . . 9
III.2 DS-TWR explanation diagram . . . 10
III.3 TDMA explanation diagram . . . 11
III.4 Ongoing discrete Kalman filter cycle (Welch and Bishop (2006, p. 5) [5]) . . . 14
III.5 Scheme of the multi-robot system and all onboard sensors" [12] . . . 16

IV.1 Configuration schema of the face to face verticaly . . . 20
IV.2 Measure for face to face verticaly referenced at 120cm . . . 20
IV.3 Mesures pour module face à face avec une référence posé à 130cm . . . 20
IV.4 Configuration schema of the face to face horizontaly . . . 21
IV.5 Schema of the paralel configuration . . . 21
IV.6 Schema of the rayon configuration . . . 22
IV.7 Schema of the face to back configuration . . . 22
IV.8 Diagram of the error based on real distance . . . 23
IV.9 Code for polynomial filter . . . 23
IV.10 Code for NLOS Filter . . . 24
IV.11 Code for innovation gate filter . . . 24
IV.12 Code for send messsage using ESP-NOW . . . 25
IV.13 Drone communication package . . . 25
IV.14 Drone communication package . . . 26
IV.15 Visualization architecture . . . 28
IV.16 Simulation architecture . . . 30
IV.17 Architecture of the first level of simulation . . . 31
IV.18 Drone capability at level 1 . . . 31
IV.19 Architecture of the second level of simulation . . . 32
IV.20 Architecture of the third level of simulation . . . 32
IV.21 Architecture of the fourth level of simulation . . . 33

# Introduction

When technological advancements on drones are growing exponentially particularly for defence usages, the study of drone swarms is more relevant than ever. Their applications are numerous : it goes from monitoring to artistic performances and rescuing. An interesting ability of drone swarms can be to carry heavy loads by coordinating the movement of multiple drones. Our project therefore lies within that application of utilizing drone swarms for carrying loads. It aims to implement drone swarms techniques based on Ultrawide Band technology (UWB) and Xbee modules.
The first objective of the project is to get familiar with the hardware available and then work out algorithms to make multiple drones fly collectively. Two main questions emerge from reflecting on the implementation of a drone swarm. Firstly, to behave collectively, each drone has to be aware of its positioning relatively to the others (or at least the neighbouring drones). Moreover, it is necessary for them to communicate with each other. Therefore, a technological choice has to be made to fulfil these two requirements. To do so, we have been imposed to work with two technologies (UWB and Xbee) and therefore, we will study how they can be helpful for responding to these problematic points.
This midterm report will cover all the research and work done since the beginning of the project in September. We will first present the hardware used as well as the specifications of the project. The bibliographical study will then be exposed followed by the current state of the work done, covering UWB, measure processing, inter-module communication, data visualisation, simulations and future tests.

# I - Hardware

## 1 - Drone specifications

In order to carry out this project, we use the quad-copter drones built at INSA Strasbourg. It is primordial to understand how drones work and what they are made of to successfully achieve our end goal.

The acronym UAV for *Unmanned Aerial Vehicle* defines an aircraft with no humans onboard and which therefore either autonomous or remotely controlled. Basically, it is a powered aerial vehicle that uses aerodynamic forces to lift itself and fly. There are many types of drones and as many different applications but some components are present in almost all of them. We can divide them roughly into two categories : mechanical and electrical.

## 2 - Drone mechanical components

### A - Structure

A drone's mechanical structure is essentially made of three parts. The frame is the skeleton of the drone and what supports all the elements. It needs to be resistant to withstand shocks but also light enough to fly longer. Carbon fibre is a commonly used material for drone frames due to its resistance and lightweight even though plastic is also used to limit costs. The size of the frame is measured from one corner to the other in a diagonal and be used to categorize drones. The *arms* link the main body part to the motors, placed at the extremities. They should robust enough to support the thrust force created by the propellers. Most drones are quad-copters with 4 arms and motors to have a good compromise between weight and manoeuvrability (6 and 8-motor drones also exist). Finally, the landing gear allows the drone to be land safely and not be damaged. They help absorb the shocks but also preserve the components from dusts when it is static on the ground. They can be fixed or folding and should also benefit from the right choice of material.

### B - Propulsion system

Electric motors are necessary to generate lift with propellers and therefore to make the drone fly. The most common type of motors on drones are 3-phased DC brushless motors which offer a high power to weight ratio while preventing heat or mechanical losses normally induced by brushes. An important parameter to consider when choosing a motor is its KV which defines the number of RPM per ESC power supply volt. The rotor size can also be seen on motors noted as "xxyy" where "xx" is the diameter of the rotor and "yy" its height.

The propellers can be chosen depending on 3 main factors. First of all, one should be cautious because props either made to turn clockwise (CW) or counter-clockwise (CCW). Secondly, the diameter and the pitch of the blade will influence the quantity of air moved by it. Therefore, it is importance to choose wisely depending on the size of the frame of the drone as well as its category (long-range, filming, racing..). Finally, the number of blades' choice has to be made according to a tradeoff between thrust and efficiency. Three-blade propellers are the most common as they provide a good ratio between trust and efficiency.

# 3 - Drone electrical system

## A - Flight controller

The flight controller or FC is the "brain" of the drone : it receives data from the different onboard sensors and send commands to the motors through the ESCs to maintain the drone's stability. It usually features a 32-bit microcontroller and sensors such as 3-axis gyro-meter and accelerometer. Other optional sensors can be connected to it like a GNSS, a magnetometer or a barometer. It is also linked to a RC receiver to be controlled remotely using high-frequency radio-communication. Common flight controller firmwares include Betaflight, Ardupilot and PX4.

## B - ESC (Electronic Speed Controllers)

ESCs may not be the first components that come to mind when thinking about drones but they are essential for its proper functioning. They serve as bridges between the flight controller's commands and the motors : they help converting the DC power from the battery to timed AC signals to control the brushless motors. It uses a MCU to decode the information from the FC and a MOSFET 3-phase inverter to ensure the DC-AC conversion. Each motor require one ESC even if today, 4-1 ESC single boards exist.

## C - Power supply

On most drones, the only power supply available is a battery. The choice of the battery is paramount because its capacity will dictate the time of flight of the drone. The main chemical technology for drone batteries is Lithium-Polymer (LiPo). Its two important parameters are its capacity C measured in mAh as well as its peak current measured in multiple of C. The number of S indicates the number of LiPo cells in series with one cell having a maximum charge of 4.2 V.

# II - Specifications

## 1 - Objective

The purpose of this project is to fly a swarm composed of 3 drone and try to lift off a payload using a net.

## 2 - Functions

- UWB precise location system
- PID Controller for drone stabilization
- ESP-NOW communication
- Go back and forth keeping the drone formation
- Software in the loop simulation
- Visualization tools for test

## 3 - Deliverables

- Report with all progress, advancements and explanation of technical knowledge
- Functional UWB code
- Methodology and results of calibration of UWB sensors
- Bibliographical Studies
- Video of drone demo
- Functional PID Corrector (C/C++)
- Functional ESP-NOW communication
- Functional ESP32 code implementation (C/C++)
- Full simulation environement based on Ardupilot and Mission Planer (Python)
- Functional visualisation tools (Matlab)

# III - Bibliographical Studies

## 1 - UWB

### A - Definitions

Ultra-Wideband (UWB) is a technology using radio frequency to exchange data. Communication is not the only purpose; UWB can also be used to locate, which is only possible by using different modules. These algorithms and techniques will be treated in the next part (B). In the majority of cases, UWB is deployed for indoor purposes; otherwise, it can be used for outdoor purposes, but this is less present in scientific papers.

### B - Location algorithm

In order to use UWB as a location sensor, we have to use specific algorithms. In the following subsections, we will explain and show 2 different location algorithms: firstly, two-way ranging (TWR), and secondly, double-sided two-way ranging (DS-TWR). Finally, we will describe TDOA, the most advanced algorithm.

#### Two way ranging (TWR)

The Two-Way Ranging (TWR) algorithm is the easiest and the fastest to implement in a UWB system. In order to explain this algorithm, we will use 2 modules called $UWB_1$ and $UWB_2$.

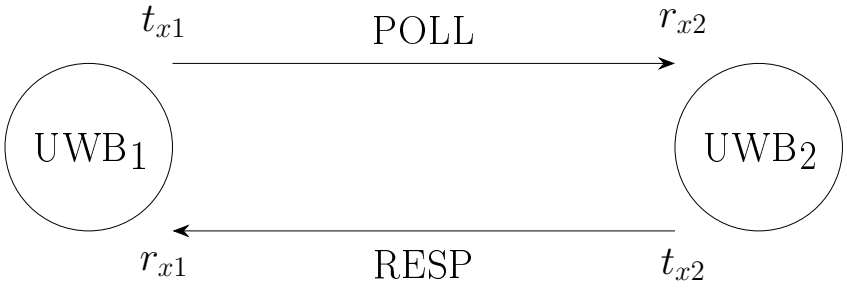



Figure III.1: TWR explanation diagram

The algorithm works in 4 steps:

1. $UWB_1$ sends a **POLL** to $UWB_2$. The time of emission $t_{x1}$ is memorized by $UWB_1$.
2. $UWB_2$ receives the **POLL** and memorizes the time of arrival $r_{x2}$.
3. $UWB_2$ processes the **POLL** and precisely plans the time of emission $t_{x2}$ for the **RESP**. It contains $(t_{x2}, r_{x2})$.
4. $UWB_1$ receives the **RESP** and memorizes $r_{x1}$.

With all these parameters, we can deduce these variables:

- Total time: $t_{tot} = r_{x1} - t_{x1}$
- Processing time of $UWB_2$: $t_{proc} = r_{x2} - t_{x2}$
- Travel time: $t_{travel} = t_{tot} - t_{proc}$

The signal travels back and forth, so it covers twice the distance ($d$), and the travel speed of a radio signal is equivalent to the speed of light ($c$).
We can deduce this equation:

$$\boxed{\frac{t_{travel} \times c}{2} = \frac{(t_{tot} - t_{proc}) \times c}{2} = \frac{((r_{x1} - t_{x1}) - (r_{x2} - t_{x2})) \times c}{2}} \tag{III.1}$$

We can also determine the error induced by this algorithm.
After calculation (B), we obtain this equation:

$$\epsilon \approx \frac{t_{proc}(e_a - e_b)}{2} \tag{III.2}$$

As we can observe, the processing time is directly linked to the error. The $t_{proc}$ is around 3,000,000 nanoseconds, so it induces a high error.

### Double Sided Two way ranging (DS-TWR)

In the TWR algorithm, the main inconvenient is the error and misprecision. To erase that error, the DS-TWR has been thinked and proposed in this paper [1]. This algorithm is also using $UWB_1$ and $UWB_2$.

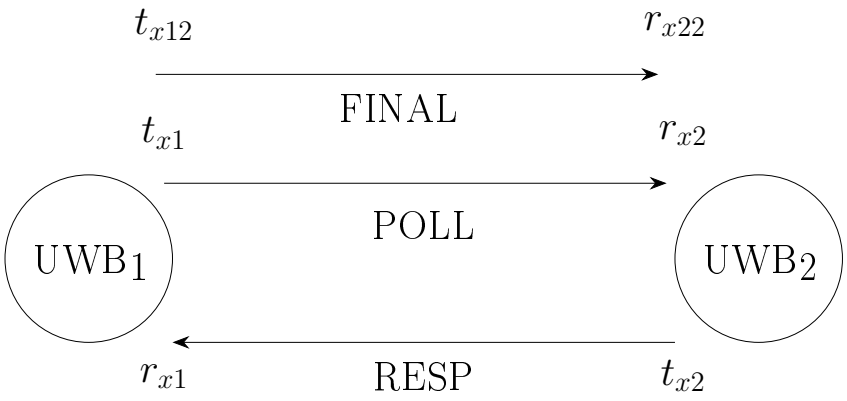



Figure III.2: DS-TWR explanation diagram

The algorithm works in 6 steps :

1. $UWB_1$ sends a **POLL** to $UWB_2$. The time of emission $t_{x1}$ is memorized by $UWB_1$.
2. $UWB_2$ receives the **POLL** and memorizes the time of arrival $r_{x2}$.
3. $UWB_2$ processes the **POLL** and precisely plans the time of emission $t_{x2}$ of the **RESP**, so that the reply delay $D_b = t_{x2} - r_{x2}$ is known.
4. $UWB_1$ receives the **RESP** and memorizes the time of arrival $r_{x1}$.
5. $UWB_1$ processes the **RESP** and precisely plans the time of emission $t_{x12}$ of the **FINAL**, so that the reply delay $D_a = t_{x12} - r_{x1}$ is known. The **FINAL** contains ($t_{x1}$, $r_{x1}$, $t_{x12}$).
6. $UWB_2$ receives the **FINAL** and memorizes the time of arrival $r_{x22}$.

With all those parameters, we compute four time intervals :

- Response time for $UWB_1 : t_1 = r_{x1} - t_{x1}$
- Response time for $UWB_2 : t_2 = r_{x22} - t_{x2}$

- Processing time of $UWB_1 : t_{proc_1} = t_{x12} - r_{x1}$
- Processing time of $UWB_2 : t_{proc_2} = r_{x2} - t_{x2}$

We can deduce the travel time:

$$\boxed{t_{travel} = \frac{t_1 \times t_2 - t_{proc_1} \times t_{proc_2}}{t_1 + t_2 + t_{proc_1} + t_{proc_2}}} \tag{III.3}$$

And then find distance :

$$d = c \times t_{travel} \tag{III.4}$$

Now let's compare the induced error. After calculation (B), we obtain this equation :

$$\epsilon \approx T_f \times \frac{e_a + e_b}{2} \tag{III.5}$$

As we can see, the error is not multiplie by $t_{proc}$, so the error is reduced.
With this algorithm, we are note dependant of the processor any more. That the algorithm currently use in our project.

### C - Timing algorithm

When working on swarm drone, there is not only two UWB modules. There is multiple UWB trying to communicate. One weakness of the UWB is it sensibility to noise and interference. The way should be clear before emiting otehrwise that create colision and damaged the singal and corrupt distance estimation. For this purpose, we need to look on how to divide time to avoid collision, in our specifif case, we are working with 3 to 4 drone.

#### ALOHA

This method is named after the first ALOHAnet system, which introduced it in 1970. It provides a simple approach to channel access control without requiring any time synchronization. Instead of planning transmissions, a module sends its data packet immediately, regardless of whether the medium is currently busy or free. The primary advantage of ALOHA lies in its ease of implementation and full decentralization, allowing independent nodes or drones to join the network dynamically without prior initialization. However, its major drawback is its susceptibility to collisions caused by uncoordinated simultaneous transmissions. As demonstrated in foundational studies, the maximum theoretical throughput (channel capacity) of ALOHA is limited to approximately 18.4% [2]. This low efficiency makes it unsuitable for high-frequency localization and critical communication applications.

#### Time-division multiple access (TDMA)

This method has for idea to divide a fixed period of time in trame, and cut it in different tranches. Each tranche is for each drone. Those trames and tranches are fixed in the hardware and are stricly respected.

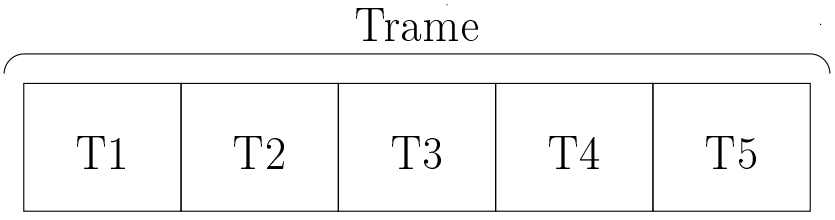



Figure III.3: TDMA explanation diagram

For exemple if we have 3 drone, trame of $UWB_1$ when is looking for distance in DS-TWR algorithm is divided by 5.

1. T1: $UWB_1$ send the **POLL** to $UWB_2$ and $UWB_3$
2. T2: $UWB_2$ send the **RESP** to $UWB_1$
3. T3: $UWB_1$ send the **FINAL** to $UWB_2$
4. T4: $UWB_3$ send the **RESP** to $UWB_1$
5. T5: $UWB_1$ send the **FINAL** to $UWB_3$

This method has an advantage: there is no collision at all due to strict time period. And also permit to shut down the module when we do not need it. But there is inconvenient, firstly if we hade one more drone, we have to add two other tranches, when we reach 10 drone, there is no time anymore to get information at a high frequency. And, it needs a synchronization to make all clock working at same speed and avoid collision perfectly.

### Slotted ALOHA

This method combines the features of both the ALOHA and TDMA protocols. Time is divided into tranches, and UWB modules are only allowed to start their transmissions at the beginning of a slot boundary. If a module has data ready to transmit, but a slot has already started, it must wait for the next available slot.

This approach preserves the flexibility of ALOHA while doubling the maximum theoretical throughput to approximately 36.8% [3]. However, it still remains less deterministic and efficient than TDMA due to potential collisions when multiple nodes choose the same slot.

# 2 - Measure processing

In most cases, data measured by sensors are unclear. It could be a sensor offset, or it can also come from crosstalk or other exterior interactions. In order to obtain clean measurements and to be able to use them for controlling or any other purpose, we have to develop some filters. In this section, we will describe three filters used for cleaning the UWB signal. Firstly, we will explain the polynomial filter, secondly the Kalman filter, and finally the extended Kalman filter.

## A - Extended Kalman Filter for relative localisation

In the context of navigating systems in noisy environments with sensors, the Kalman filter is a relevant algorithm. As Welch and Bishop (2006, p. 1) [3] point out, the Kalman filter is notable for its ability to "support estimations of past, present, and even future states [...] even when the precise nature of the modeled system is unknown.". In our study case, we deal with a drone swarm communicating using the UWB communication protocol, which gives us the relative distance between each pair of drones by using the time of flight of a radio wave. The UWB therefore provides a scalar distance, whereas the quantity of interest is the relative position vector. The measurement model, which links the two through the Euclidean norm, is nonlinear, which calls for the extended Kalman filter. This version resorts to a typical technique of linearization around the current state using a Jacobian matrix. We are also considering an anchor-free system which enhances the challenge of this project. In this section, we will first cover the theory of the classical Kalman filter and its mathematical formalist and then dive into the extended Kalman filter and its utility.

### Theory of the Kalman filter

**Origin of the theory** In his pioneering article of 1960 [4], Rudolf Kalman describes state estimation as a recursive process in which at each time step, the current estimation and the uncertainty are computed using only the past estimate and the newly available measure. When assuming a linear dynamical system and Gaussian noise, the Kalman filter minimizes the mean-square estimation error : in other words, the best possible estimate given the available measurements. Even when these assumptions do not hold exactly, the filter remains the best linear estimator achievable (known as the Best Linear Unbiased Estimator, or BLUE). This optimality is what established the Kalman filter as the standard approach for state estimation in noisy dynamic systems. While Kalman filter's optimality holds for linear dynamical systems, we have in our case, as mentioned earlier, a non-linearity with the use of the Euclidian distance. This is why we require an extended version of the filter that can handle the non-linearity, even at the cost of losing the strict optimality.

**Formalism of the discrete Kalman Filter** Let's note $x_k \in \mathbb{R}_n$ the current state of a discrete system governed by the following linear equation :

$$\mathbf{x}_k = \mathbf{F}_k \mathbf{x}_{k-1} + \mathbf{B}_k \mathbf{u}_k + \mathbf{w}_k$$

In which we have :

- $F_k$ is the state transition matrix, linking the previous state $x_{k-1}$ to the current state $x_k$
- $B_k$ is the control-input matrix, linking the control vector $u_k$ to the current state $x_k$
- $w_k$ is the process noise which is assumed to be Gaussian and centered, with a covariance matrix $Q_k$

At each time $k$, the measurement $z_k$ is written as follows :

$$\mathbf{z}_k = \mathbf{H}_k\mathbf{x}_k + \mathbf{v}_k$$

With :

- $H_k$ is the observation matrix
- $w_k$ is the observation noise which is assumed to be Gaussian and centered, with a covariance matrix $R_k$

We can divide the Kalman filter process into 2 phases : the "prediction" phase and the "update" phase. The first one does an estimate of the state at the next time step by using the estimate of the current time step and the error covariance. The second phase can be thought of as a feedback : it includes a new measurement in the current state to improve its estimate.

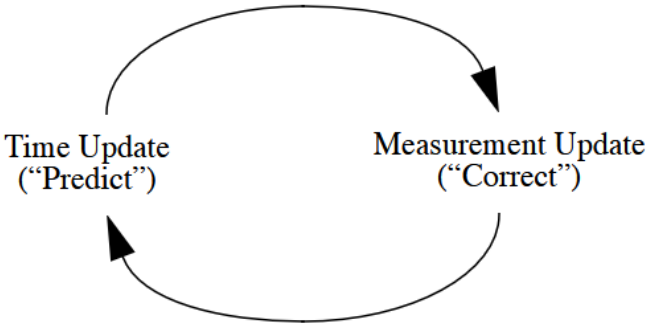



Figure III.4: Ongoing discrete Kalman filter cycle (Welch and Bishop (2006, p. 5) [5])

### Extended Kalman filter

As mentioned earlier, the extended Kalman filter is used to deal with a non-linear state process defined as diffentiable function $f$. To do so, it essentially linearizes the nonlinear function around the current estimate by computing the Jacobian matrices of the nonlinear functions $f$ and $h$ in the following equations (with the same notations as before) :

$$\mathbf{x}_k = f(\mathbf{x}_{k-1}, \mathbf{u}_k, \mathbf{w}_k)$$

$$\mathbf{z}_k = h(\mathbf{x}_k, \mathbf{v}_k)$$

Like the original filter, it is recursive: only the current estimate and its covariance are stored, which keeps memory and computation low enough for embedded platforms, while the covariance provides a measure of uncertainty that can be used to reject inconsistent measurements. However, this comes with a drawback. The linearization is a first-order approximation, so the optimality of the linear Gaussian case is lost, and the filter may diverge if the initial estimate is poor or the system is strongly nonlinear [6]. Its performance also depends on the tuning of the noise covariances. Alternatives such as the unscented Kalman filter [7], particle filters [8] or optimization-based estimators differ in accuracy, robustness and computational cost. The EKF nevertheless remains the standard approach in navigation, including in open-source flight controllers (like in the ArduPilot firmware).

## B - UWB localization : benefits and limits

As described in the previous section, UWB ranging enables centimeter-level accuracy via time of flight calculations. However, the data gives us only the relative distances between each drone. This data lacks information, particularly about orientation. In this section, we will discuss the viability of a distance-only approach and the topic of observability.

### Range-only system

In their paper entitled "Dynamic Range-Only Localization for Multi-Robot Systems", Cao et al. (2018) [9] propose a method to perform robot localization via range measurement only. To do so, they present a technique based on successive stages which allows to dynamically localize multiple robots by using trilateration and an EKF recursive estimation. While this article shows a functional multi-agent system with relative localization without anchors, it has been demonstrated only in simulation and in a 2D environment : it does not address any experimental validation nor three-dimensional flight. Moreover, their approach uses a coordinate system built by the robots themselves while drones are usually localized in an Earth-fixed frame via their flight controllers. The authors do acknowledge the question of a global frame but consider it outside the scope of their work, mentioning that a swarm deployed in an unknown environment might no have access to global positioning.

### Observability

As defined in [10], we say that "a system is *observable* if for every $T > 0$, it is possible to determine the state of the system $x(T)$ through measurements of the $y(t)$ and $u(t)$ in the interval $[0, T]$" where $y(t)$ and $u(t)$ are respectively the output and the input of the process at time $t$. Observability plays an important role in control theory because it is common that all the states are not accessible through measurement : it is therefore possible to build an *observer*, a dynamical system represented by a mathematical model capable of recreating the states of the system.

In our project, the observability problem is harder to deal with because it would be necessary to know which states of the drones we choose to consider. To get a rough idea of the problem, we can look at the article from Van der helm et al. (2020) [11] who studed a similar problem : they implemented UltraWide band technology for both ranging and communication between Parrot drones (transmission of the velocities, accelerations, yaw rates and heights). They tested different configurations of leader-follower flights and performed observability analyses. They showed, on the one hand, that without a common heading, "the motion of agents must meet more stringent conditions to be observable and the relative localization becomes more susceptible to noise on the range measurements". On the other hand, the dependency on a common heading offers the advantage of being less sensitive to local disturbances in Earth's magnetic field, which, even when small, can lead to a large negative impact on the EFK used. In a leader-follower configuration, a fixed-geometry (like 2 drones moving in parallel) is not possible and for that, they implement a method where the follower flies in a time-delayed trajectory version of the leader. Their approach was validated indoors with an on-board EKF. Howewer outdoor operations with GNSS (Global Navigation Satellite System) are not addressed.

## C - Data fusion

### Data fusion with an EFK in drone swarm applications

In the context of our project, the most relevant article is most likely "Onboard Ranging-Based Relative Localization and Stability for Lightweight Aerial Swarms" by Li et al. (2025) [12] from TU Delft in the Netherlands. In their paper, they share an autonomous relative localization

technique fusing UWB distance measurements and state information like velocity, yaw rate, height from neighbours. The following illustration shows their scheme of work :

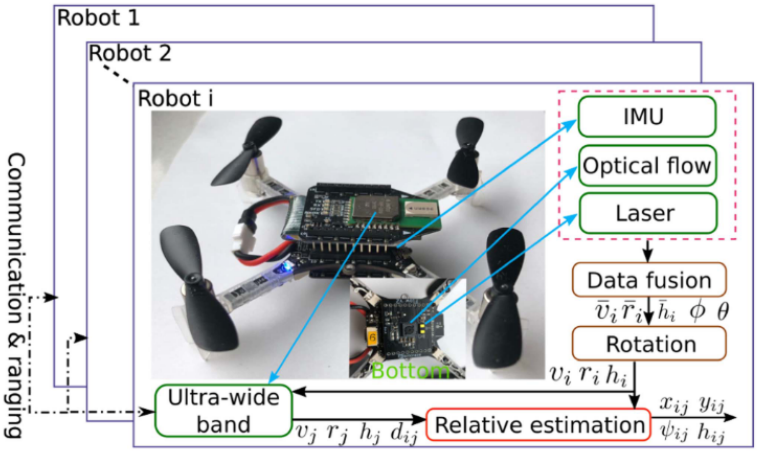



Figure III.5: Scheme of the multi-robot system and all onboard sensors" [12]

By using the data of 3 sensors (IMU, optical flow, laser), they retrieve the 3-axis velocity $\overline{v_i}$, the pitch $\theta$ and the roll $\phi$ in the body frame of the drone. The yaw rate $\overline{r}$, the range $d$ (distance between 2 drones) and the height $h$ are respectively given by the gyroscope, the UWB sensors and the laser. One of the main aspects of their project is that they work with tiny, lightweight (33 grams) drones which are resource-constrained. Therefore, an Extended Kalman filter is exploited to combine the data, particularly because it has a low computational overhead. The article is particulary appropriate because it deals with every aspect of our project. It brings up the fusion of multiple sensors through an EFK, the use of Ultra-wide band for both ranging and communication as well as observability considerations and hardware resource constraints. Nevertheless, they do not implement GPS when merging data, which differs from the context of our work.

### GPS consideration

The papers mentioned previously did not include GPS localization in their architecture but our drones do include such a tracker, which we would like to take advantage of. We are working on an outdoor system and it is possible through the flight controller to use the GPS data to get a more accurate state estimate of the drone. The details about the hardware will be given in the next section.

In an article of 2020, Chen et al. [13] studied the complementarity of UWB measurements with GPS localization to get a better accuracy. They showed a great improvement of devices localization with an accuracy up to 10 centimeters. This approach is different from ours as the GPS data is directly combined through a cost function with the UWB measurements while we are planning to use an Extended Kalman filter. Additionally, their experiments were made on static objects and the final localization is obtained in a 2D plane, while we deal with moving UAVs in a 3D frame, which may add more challenges.

A closer reasearch to our situation was made by Yang et Lee in 2023 [14] in which they implemented an Extended Kalman Filter using GNSS and INS (Inertial Navigation System) with complementary UWB measurements in a flight controller firmware. They used the flight controller "Pixhawk 4" (the same as us) and successfully carried out a field test which, however, did not include a moving drone formation.

# 3 - Communication protocol

## A - Xbee

XBee is a family of radio modules made by Digi International. Depending on the model and firmware, it implements IEEE 802.15.4 [15] directly, ZigBee, or Digi's own DigiMesh protocol. The module is controlled by a microcontroller through a simple serial link (UART or SPI), either in transparent mode (the module behaves like a wireless serial cable) or in API mode (the host sends and receives structured frames with addresses and acknowledgements).

Two bands are commonly used. The 2.4 GHz modules have a high data rate and are widely available. The 900 MHz modules trade throughput for a much longer range. At the MAC layer, 802.15.4 [15] uses CSMA/CA, so a module listens before transmitting, and unicast frames can be acknowledged and retried automatically.

| **Criterion** | **XBee** |
|---|---|
| Range | $\sim$ 60 m indoor, $\sim$1 km outdoor line-of-sight |
| Latency | A few ms to tens of ms per hop |
| Data rate | 250 kbps raw |
| Power consumption | $\sim$ 30–45 mA in RX/TX (standard module) |
| Topology | Point-to-point, star, mesh (ZigBee, DigiMesh) |
| Number of nodes | Hundreds (16-bit addressing) |

Table III.1: XBee characteristics

## B - UWB

Besides ranging, a UWB module is a radio transceiver, so the frames used for ranging (POLL, RESP, FINAL) can carry a small payload. In the DS-TWR algorithm described above, the FINAL frame already contains timestamps (tx1, rx1, tx12). The same mechanism can carry a drone ID, a status flag or a short command. The advantage is that localization and data exchange share a single radio and a single time schedule, so no second module is needed for small messages.

The main limitation is the frame size and the energy cost. UWB frames are short (about 127 bytes in the standard configuration), and the receiver is power-hungry compared with the other technologies in this section. UWB is therefore well suited to short control messages, but not to image or bulk data transfer.

| **Criterion** | **UWB** |
|---|---|
| Range | $\sim$10–50 m in practice indoors, up to $\sim$300 m line-of-sight |
| Latency | Frame airtime under 1 ms; a full DS-TWR exchange takes a few ms |
| Data rate | 110 kbps, 850 kbps or 6.8 Mbps; small payload per frame ($\sim$127 bytes) |
| Power consumption | $\sim$ 30–40 mA in TX; RX is higher |
| Topology | Peer-to-peer / star (tags and anchors); no native mesh |
| Number of nodes | Limited by the channel access method (ALOHA, TDMA); a few tens with ranging |

Table III.2: UWB characteristics

### C - ESP-NOW

ESP-NOW is a protocol developed by Espressif for its ESP32 and ESP8266 chips. It uses Wi-Fi radio hardware at 2.4 GHz, but without a router, an access point or an IP connection: data is sent in short vendor-specific Wi-Fi frames directly from one device to another, identified by their MAC addresses. A device registers its peers, sends a packet (up to 250 bytes in the classic version), and receives a callback telling it whether the frame was acknowledged. Broadcast is also possible, and encryption can be enabled per peer. The ESP-NOW connection is connectionless, so there is no handshake before sending. This explains the low latency.

Studies agree on its main features [16]. Authors compared ESP-NOW with Wi-Fi and Bluetooth on range, speed, latency, energy usage and barrier resistance. And evaluated the maximum distance and power consumption of ESP-NOW nodes. A community test [17] published by Espressif reports latency under 20 ms in open field up to 300 m, and a long-range mode (ESP-NOW-LR) reaching up to about a kilometre. Note that this source is not peer-reviewed.

| **Criterion** | **ESP-NOW** |
|---|---|
| Range | $\sim$ 15–50 m indoors; $\sim$ 200–300 m outdoors line-of-sight; up to $\sim$1 km |
| Latency | Typically a few ms; under $\sim$ 20 ms measured in open field |
| Data rate | Depends on the configured Wi-Fi rate (1 Mbps by default), 250 bytes per packet |
| Power consumption | $\sim$ 100–240 mA in RX/TX; |
| Topology | Peer-to-peer, one-to-many, broadcast |
| Number of nodes | About 20 registered peers per device; broadcast has no such limit |

Table III.3: ESP-NOW characteristics

# IV - Design and Implementation

## 1 - UWB

In swarm drone project, one information is central and will determine the achievement of the shaping. This information is the **position**. In our case, we are measuring the relative position between drone.
To achieve this needed, our professor imposed UWB as localisation sensors. Those sensors are based on electromagnetic signal and needed two modules to interact with each others (Find more information in the bibliographic studies III.

### A - Algorithm choice

As expose in the bibliographic studies, there are differents algorithms which respond to multiples needs. In this project, we need precise localisation data and no interference, furthermore we have maximun 3 drone in the swarm.
With those caracterestics, we choose the DS-TWR location algorithm (B) and the TDMA timing algorithm (C).
We have implemented them into the ESP32-DW1000 module.The *code* is using C/C++ as software language.

### B - Calibration

In general, sensors need a calibration to use them efficiently. For UWB module, we need to establish a distance reference. For this project, the reference is the expected distance between drones: 1.20m. Afterward, we had to adapt the Antenna Delay parameters which represent signal's time in antenna's PCB. Each module has different time delay. To calibrate, we had to choose a module as a reference, here UWB 1. And then modify Antenna Delay of each drone to make them find 1.20m at 1.20m in real life. After this step, all drones are calibrated for each other. After doing some tests, we had to modify the reference to 1.30m, this modification is explained in the next part C.

### C - Test

During the last year project, our colleagues find out that UWB module should be vertically installed to optimize there precision and reach. So we have repeated the same test in order to verify there result.

#### Face to Face verticaly

We have carried out same vertical test following this schema :

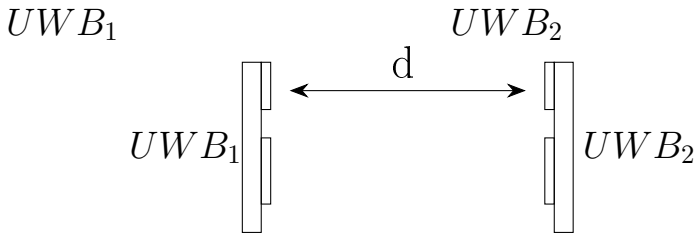



Figure IV.1: Configuration schema of the face to face verticaly

| Real distance (cm) | Measured distance (cm) | Error |
|---|---|---|
| 70 | 67 | 3 |
| 80 | 78 | 2 |
| 90 | 89,2 | 0,8 |
| 100 | 100 | 0 |
| 110 | 110,6 | -0,6 |
| 120 | 120 | 0 |
| 130 | 132,4 | -2,4 |
| 140 | 143,6 | -3,6 |
| 150 | 154,2 | -4,2 |
| 160 | 165,7 | -5,7 |
| 170 | 176,4 | -6,4 |

Figure IV.2: Measure for face to face verticaly referenced at 120cm

| Real distance (cm) | Measured distance (cm) | Error |
|---|---|---|
| 70 | 64 | 6 |
| 80 | 76,2 | 3,8 |
| 90 | 85,6 | 4,4 |
| 100 | 97,4 | 2,6 |
| 110 | 108,2 | 1,8 |
| 120 | 119,6 | 0,4 |
| 130 | 130 | 0 |
| 140 | 140 | 0 |
| 150 | 152,7 | -2,7 |
| 160 | 162 | -2 |
| 170 | 172,2 | -2,2 |

Figure IV.3: Mesures pour module face à face avec une référence posé à 130cm

As we can observe in those table, the precision is decreasing fast when distance is higher than the reference. So place the reference at 130cm would allow us to get better precision at $\pm$ 10cm around 120cm.

### Face to Face horizontaly

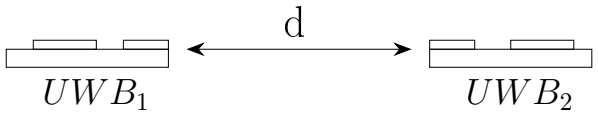



Figure IV.4: Configuration schema of the face to face horizontaly

| Real distance (cm) | Measured distance (cm) | Error |
|---|---|---|
| 69,4 | 61 | -8,4 |
| 70 | 62,2 | -7,8 |
| 75 | 69,6 | -5,4 |
| 80 | 73,9 | -6,1 |
| 82,6 | 76,1 | -6,5 |
| 85 | 79,9 | -5,1 |
| 90 | 86,6 | -3,4 |
| 91 | 87 | -4 |
| 95 | 92,2 | -2,8 |
| 100 | 96,5 | -3,5 |
| 100 | 96,8 | -3,2 |
| 105 | 102,6 | -2,4 |
| 107,3 | 106,4 | -0,9 |
| 110 | 108,9 | -1,1 |
| 115 | 113,4 | -1,6 |
| 120,5 | 120,9 | 0,4 |
| 125 | 126,4 | 1,4 |
| 130 | 129,6 | -0,4 |
| 132 | 133,7 | 1,7 |

| Real distance (cm) | Measured distance (cm) | Error |
|---|---|---|
| 135 | 133,6 | -1,4 |
| 140 | 138,6 | -1,4 |
| 145 | 144,9 | -0,1 |
| 147,4 | 150 | 2,6 |
| 150 | 151,7 | 1,7 |
| 155 | 160,3 | 5,3 |
| 155,5 | 161 | 5,5 |
| 160 | 165,7 | 5,7 |
| 164,7 | 171,8 | 7,1 |
| 165 | 172,2 | 7,2 |
| 167 | 172,8 | 5,8 |
| 170 | 177,2 | 7,2 |
| 175 | 180,3 | 5,3 |
| 176 | 183,8 | 7,8 |
| 180 | 184,3 | 4,3 |
| 185 | 188,3 | 3,3 |
| 189,6 | 197,9 | 8,3 |
| 190 | 191,9 | 1,9 |

Table IV.1: Mesure face à face de façon horizontale

Let's compare both position (horizontal or vertical), as we can observe around 120cm for horizontal position, we got around 1.8cm of error and for vertical, we got around 1cm. Also, when the distance move off 120 cm, the precision decreases slower in vertical than in horizontal. Based on this experimentation, we choose to use our module verticaly to get better precision and flexibility.

Normally, in the documentation, there is an antenna diagram which describe the efficient area of the antenna. We were looking for this diagram in order to choose the shape of the drone which get the best precision. After research, we did not find it, therefore we had to realize some tests in different disposition.

### Paralel module

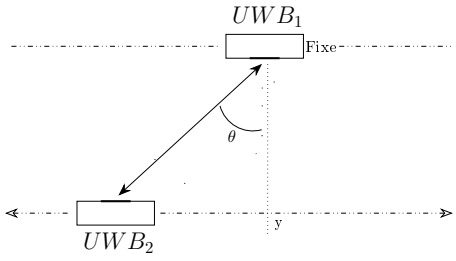



Figure IV.5: Schema of the paralel configuration

| offset (cm) | Angle (°) | Real distance (cm) | Measured distance (cm) | Error |
|---|---|---|---|---|
| 0 | 0 | 120 | 117 | 3 |
| 10 | 4,78 | 121 | 118,4 | 2,6 |
| 20 | 9,59 | 121,5 | 118,9 | 2,6 |
| 30 | 14,48 | 122,5 | 119,1 | 3,4 |
| 40 | 19,47 | 125,8 | 124,8 | 1 |
| 50 | 24,62 | 128,5 | 128,7 | -0,2 |
| 60 | 30,00 | 132,5 | 134,4 | -1,9 |
| 70 | 35,69 | 137 | 142,6 | -5,6 |

Table IV.2: Measure of the paralel configuration

### Rayon configuration

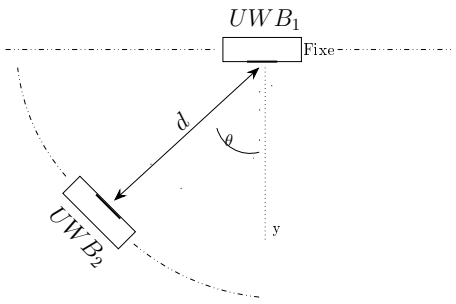



Figure IV.6: Schema of the rayon configuration

| Angle(°) | Real distance (cm) | Measured distance (cm) |
|---|---|---|
| 10 | 120 | 116,7 |
| 20 | 120 | 118,4 |
| 30 | 120 | 119,5 |
| 40 | 120 | 122,8 |
| 50 | 120 | 125,9 |
| 60 | 120 | 124,2 |
| 70 | 120 | 123 |
| 80 | 120 | 124 |
| 90 | 120 | 130 |

Table IV.3: Measure of the rayon configuration

### Face to Back

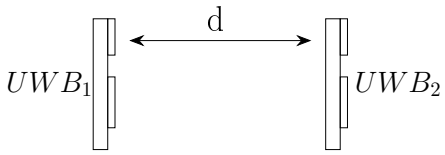



Figure IV.7: Schema of the face to back configuration

| Real distance (cm) | Measured distance (cm) | Error |
|---|---|---|
| 70 | 97 | -27 |
| 80 | 109,8 | -29,8 |
| 90 | 122,8 | -32,8 |
| 100 | 133,3 | -33,3 |
| 110 | 143,5 | -33,5 |
| 120 | 152 | -32 |
| 130 | 161,6 | -31,6 |
| 140 | 177,2 | -37,2 |
| 150 | 188 | -38 |
| 160 | 192,2 | -32,2 |
| 170 | 208,8 | -38,8 |

Table IV.4: Measure of the face to back configuration

### D - Filter

Electromagnetics signals are sensitive to exterior noise and can be interfered easily. Therefore, we have to filter the raw data. To do so we implement three differents filters.

#### Polynomial filter

During calibration and test, we have discovered that the errors is linked to the real distance and variate following a polynomial equation.

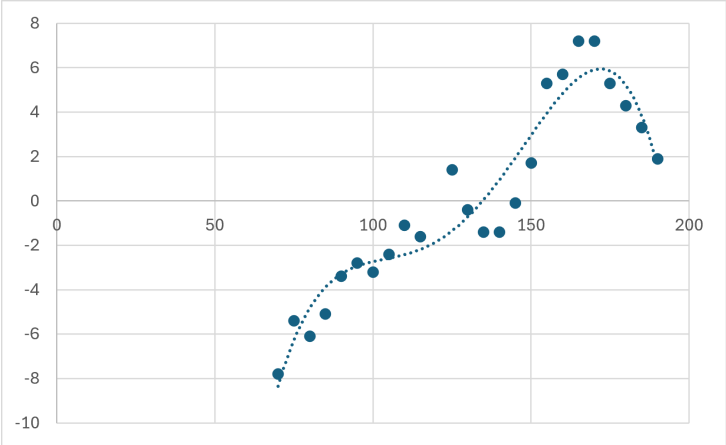



Figure IV.8: Diagram of the error based on real distance

Using the Excel linearization tool, we got this error equation:

$$f(x) = -1{,}004 \times 10^{-6}\, x^4 + 0{,}0005059\, x^3 - 0{,}09263\, x^2 + 7{,}398\, x - 221{,}7 \qquad \text{(IV.1)}$$

Every measure pass through the equation and are recalibrate based on this equation. That permit to reduce the error when we are at more ± 10 cm from the reference.

We have implemented it in C/C++ :

```c
void polynomial_filter(float* d) {
    float raw_cm = *d * 100.0f;

    // Equation d'erreur = a5*x^4 + a4x^3 + a3*x^2 + a2*x + a1
    const float a5 = -1.004e-6f;
    const float a4 = 0.0005059f;
    const float a3 = -0.09263f;
    const float a2 = 7.398f;
    const float a1 = -221.7;

    float estimated_error_cm = a1 + raw_cm * (a2 + raw_cm * (a3 + raw_cm * (a4 + raw_cm * a5)));

    float corrected_cm = raw_cm - estimated_error_cm;

    if (corrected_cm < 0.0f) {
        corrected_cm = 0.0f;
    }

    *d = corrected_cm / 100.0f;
}
```

Figure IV.9: Code for polynomial filter

### NLOS Filter

Electromagnetic signal are very sensitive to obstacle. After doing research, we have discovered that the signal will lost power if it encounters an obstacle. Based on that, we can verify that the power receives is strong enough otherwise the value cannot be considered as feasible so we can dennied it.

We have implemented it in C/C++:

```c
typedef struct {
    float firstPathPower;
    float receivePower;
    float firstPathAmp1;
} UWB_Diagnostics;

bool NLOS_filter(UWB_Diagnostics diag) {

    float min_fpp_threshold = -85.0f;
    if (diag.firstPathPower < min_fpp_threshold) {
        return false;
    }

    float power_diff = fabsf(diag.receivePower - diag.firstPathPower);
    float max_power_diff = 6.0f;

    if (power_diff > max_power_diff) {
        return false;
    }

    return true;
}
```

Figure IV.10: Code for NLOS Filter

### Innovation Gate Filter

UWB has tendencies to be noisy, that create some absurd value very high or very low. This data can affect the corrector and create some discontinuity in the drone control. Based on the max speed of the drone, we can determine absurdity by calculating the difference $d_{-1}$ and $d$, then determine if the difference can be reach at the max speed, if not the measure is ignored, otherwise, we accepted it.

We have implemented it in C/C+:

```c
bool innovationGate_filter(float current_d, float prev_d, float max_speed, float dt) {

    float max_possible_delta = max_speed * dt;

    float sensor_noise_margin = 0.2f;
    float allowed_change = max_possible_delta + sensor_noise_margin;

    float actual_change = fabsf(current_d - prev_d);

    if (actual_change > allowed_change) {
        return false;
    }

    return true;
}
```

Figure IV.11: Code for innovation gate filter

## 2 - Communication

In the project, we have to differents use for communication, first communication is used to made each drone talk with each other to maintain the shape but also adapt themselves to the other. Second communication is the data's descent from the drone to the base to permit visualition of the swarm and command drone.

### A - Choice

We choose ESP-NOW has the protocol for two main raison : it doesn't need new hardware implementation. And, it responds to both communication purpose of the project. Then, has said during the presentation, the code has been optimized and write by the manufacturer which facilitate it implementation. For exemple, send data is only this command :

```
esp_now_send(MAC_ADRESS, data, lenght_of_data);
```

Figure IV.12: Code for send messsage using ESP-NOW

Has said before, range reach 150 meters allows us to create a base. It only ask us, to have a ESP32, connect to our computer.

### B - ESP-NOW implementation

To implement ESP-NOW, we had to think about the data we want to exchange. For our first test, we are using two different packages for two different purpose.
First for inter-drone communication :

```
typedef struct __attribute__((packed)) {
  uint8_t  id;
  uint32_t t_ms;
  float d;
  float z;
  GPS gps;
  Attitude a;
} Pkt;
```

Figure IV.13: Drone communication package

And for the base to drone communcation we have this package :

```c
typedef struct __attribute__((packed)) {
  uint8_t  id;
  uint32_t t_ms;
  uint8_t commande;
  float variable;
} Pkt_base;
```

Figure IV.14: Drone communication package

## 3 - Measure processing and formation control

After studying the UWB measurements and the communication with ESP-NOW, we have to consider the processing of the retrieved data to finally implement an optimal formation control algorithm. The questions that need to be asked are the following : what data do we need to retrieve and what algorithm do we use for its processing and for the formation control. As mentioned before, our drones include many onboard sensors and benefit from the added UWB module. The bibliographical study gave us insights about potential architectures of measure and control so a choice has to be made considering the hardware available. Here are two things to notice :

- UWB modules return purely relative distance measurements and give no sense of direction at all
- The onboard Extended Kalman Filter from the flight controller's firmware Ardupilot fuses the measurements from its different sensors (accelerometer, gyroscope, magnetometer and barometer) and the added GNSS module to get notably an accurate estimation of the drone's attitude (yaw, roll, pitch) as a quaternion, 3 NED (North East Down) velocity components and 3 NED position components (up to 22 states are estimated in total).

Therefore, we know that we can get good estimates of these states through the onboard EKF of the flight controller. A first choice has to be made about what we want to control.

# 4 - Visualization

In outdoor test, drones are flying at more than 10 meters away from us. That constraint reduces our visibility and comprehension of the drone movements, attitude and position. To prevent this problematic we need to develop a visualization system. That would increase our error detection capability and give the opportunity to save test and simulate them afterward to analyse and detect problematic point.
Visualization would also allow us to observe in real time differents parameters as: attitude, 3D shape positionning, and also RAW data coming from the drone.

In order to achieve this system, we choose the sofware Matlab as a visualizer and software language.

## A - Matlab

Matlab is a general tool developped by the MathWorks. It has many functions such as mathematics calculator, simulator, and also a corrector designer. Based on add-on and toolbox, it can be usefull for many differents purpose. It also permits to generate an interface.

## B - Architecture

The visualization follow the next architecture :

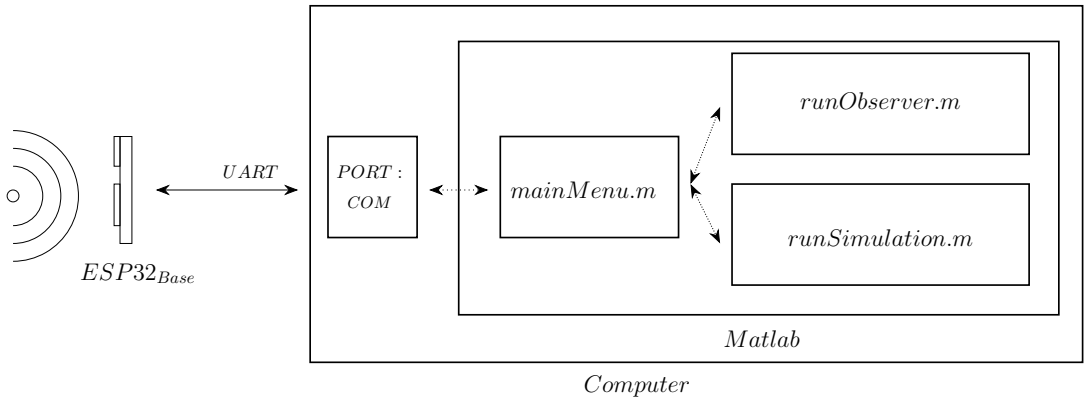



Figure IV.15: Visualization architecture

Let's explain each different bloc and its role :

- ESP32 : Relay between ESP-NOW protocol and the UART Port of the computer, which permit to make it visible for the matlab code.
- COM PORT: Physical bridge between the computer and the ESP32
- mainMenu : General window which redirect for the next window
- runObserver : Window to observe UART communication on the PORT (Debug utility)
- runSimulation : Main Window where we visualise attitude of the drone, shape placement and data reader

# 5 - Simulation

Due to logistical problem, outdoor test are rare, and INSA Strasborug does not have indoor test room. Howewer our work need to be tested and confronte reality. To achieve this purpose, we had to develop a simulation protocol which is the nearest possible from the reality.
To do so, we had to choose same firmware as our flight controler and an UAV simulation tool.
In this part, we will first present both firmware : ArduPilot and Mission planer, second, explain the simulation architecture, then present the result we obtained and finally the simulation possibilities.

## A - ArduPilot and Mission Planer

To do drone simulation, we have to select the same architecture as our drone on the pixhwak 4. In the drone industry, two differents architectures are existing: Ardupilot and PX4. In our case, pixhwak 4 is using the ArduPilot architectures.
Then, after selecting the architecture, we have to use a software to visualise and simulate drone. We choose mission planer because we have already work with it on different project.
Let's present both tools:

### ArduPilot

ArduPilot was created in 2007 by the DIY Drones community. Originally developed for Arduino boards, it has since become a general-purpose, open-source autopilot. It can control many types of vehicles (copters, planes, rovers, submarines, etc.). ArduPilot provides autopilot functions, receives data from sensors, and fuses it using an Extended Kalman Filter (EKF) [ref]. It also allows users to create pre-planned missions. To operate, it uses the MAVLink protocol to communicate with the drone and facilitate control. In our case, an ESP32 uses the MAVLink library to send MAVLink messages that command the drone's movements.

### Mission planer

Mission planer is a representative tool for copters such as drone. Created by Michael Oborne to help users of ArduPilot architecture to vizualise their drone and command them. The software also permits to declare multiple drone so create a swarm. For our project, it will be used as a simulation field.

### B - Architecture

The simulation works with a very specific architecture cause tools like ArduPilot works under Linux system. The architecture is exposed in the figure below.

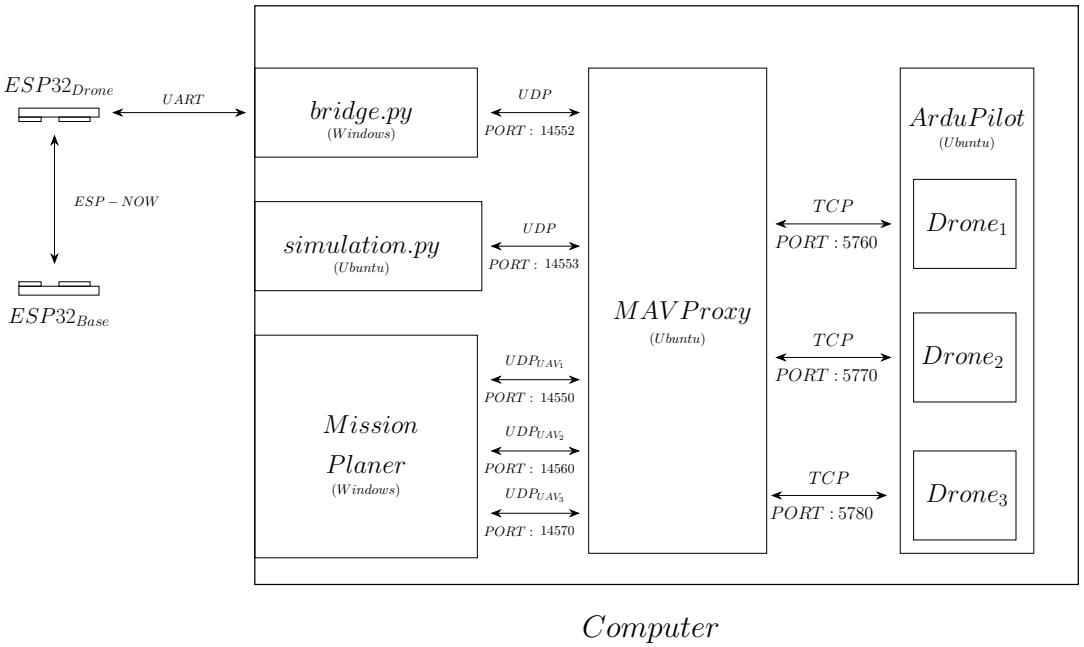



Figure IV.16: Simulation architecture

Let's explain each different bloc and its role :

- $ESP - 32_{Base}$ : Represent order send from the base for exemple takeoff
- $ESP - 32_{Drone}$: Represent the main drone where code we want to test is running
- ArduPilot : Represent the pixhawk of our drone. Here, we are simulating 3 of them.
- Bridge.py : Code made to redirect data from the UART connection to UDP connection
- MAVProxy : Manage all entries and exit permit to link the drone entries redirect them to the Ardupilot and also permit to redirect exit from ArduPilot (GPS, speed, etc) to Mission Planer.
- Mission Planer : Use to visualise our swarm drone based on data from ArduPilot
- Simulation.py : Python scrip to simulate $Drone_2$ and $Drone_3$

# 6 - Test

Test is the main part of technical project, reality is always different from the theory. Therefore, we have to carry out them through two different environement: simulation and outdoor.

## A - Simulation

Using the architecture describe in the part 5, we did and planned to do different test for different axes of the project.

### Ongoing :

Console control:
This test has for purpose to test MAVLink function, communication and drone control.
It follows the next schema:

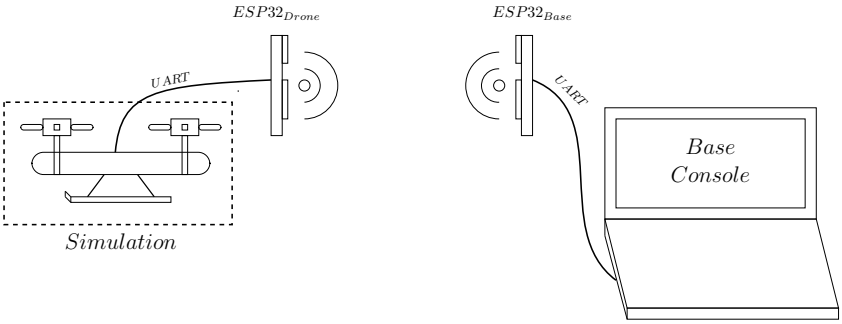



Figure IV.17: Architecture of the first level of simulation

The drone is for now capable of this function (lexicon below) :

```
[1] => status
[2] => mode
    [1] = guided
    [2] = land
    [3] = rtl
    [4] = stabilize
    [5] = loiter
    [6] = mode auto
[3] => takeoff
[4] => armement
[5] => parameters

```

Figure IV.18: Drone capability at level 1

We can conclude that our communication between base and drone, is well-designed and the MAVLink procedure is well send to the drone.

### Planned :

Level 2:
Objective of this level, a slave drone need to follow the master drone controlled by the console

as in the level 1. To do so it should use communication, attitude and GPS data. Possibly, we can think of integrate UWB to make a full test, closest to the real situation.

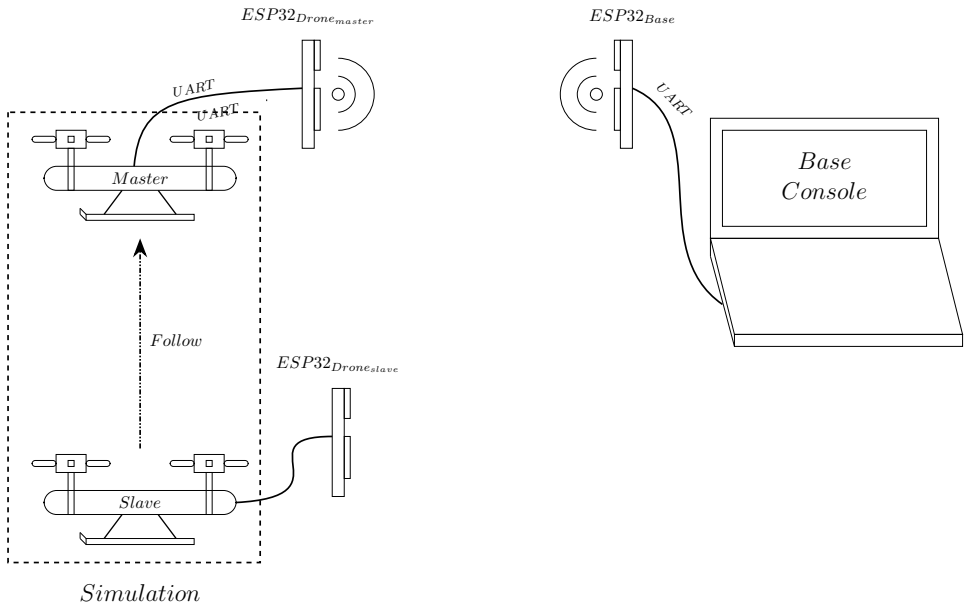



Figure IV.19: Architecture of the second level of simulation

Level 3:
Objective of this level is to reach 3 drone and introduce swarm drone. On the 3 drones, two of them would be control by a python script. The third drone is the tested one. It permits to test corrector and attitude corrector.

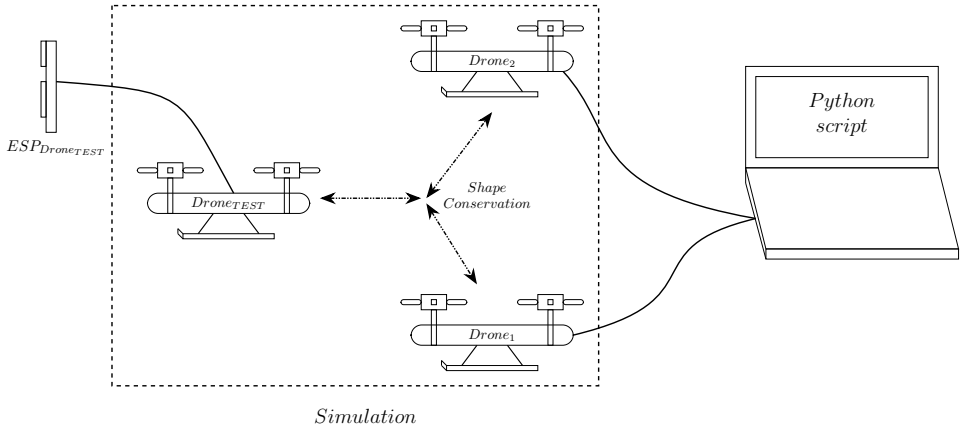



Figure IV.20: Architecture of the third level of simulation

Level 4:
Objective of this level is to reach a full simulation environement where all the three drone are working on it own ESP32. Mission planer and Ardupilot are strictly an environement simulation. It is the closest simulation to the objective of the project.

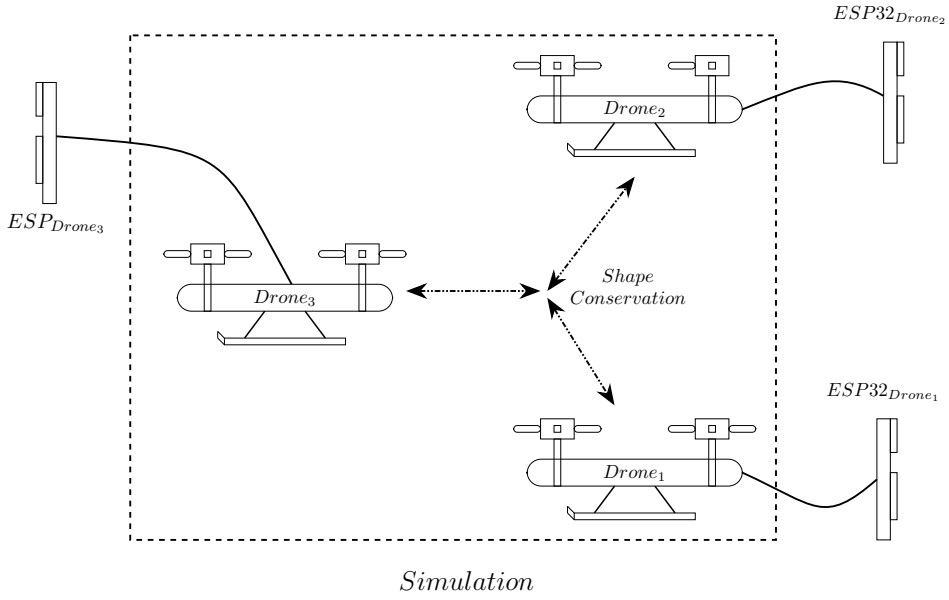



Figure IV.21: Architecture of the fourth level of simulation

### B - Outdoor

Make drone flying is restricted by the laws. It makes test difficult to proceed and rare. Therefore, outdoor test needs to be precisely prepare to not lose any time and be efficient on site.

#### Planned

We have planned to attempt outdoor test within 2 weeks. During this test, we would love to test:

- UWB placement (rotation, and heigh)
- ESP-NOW range
- Drone control console (as simulation level 1 [A])
- Slave drone following master drone (as simulation level 1 [A])

# Conclusion

# Calculations

### TWR Error calculation :

Let's set:

- True time of flight : $T_f$
- True time of process : $t_{proc}$
- Clock drift errors : $e_a$ and $e_b$

The measured times are:

$$R_a = (2T_f + t_{proc})(1 + e_a) \text{ and } t_{proc_b} = t_{proc}(1 + e_b)$$

The calculated time of flight is:

$$\hat{T}_f = \frac{R_a - t_{proc_b}}{2}$$

By replacing $R_a$ and $t_{proc_b}$, we obtain :

$$\hat{T}_f = \frac{2T_f(1 + e_a) + t_{proc}(1 + e_a) - t_{proc}(1 + e_b)}{2}$$

By simplifying, we obtain :

$$\hat{T}_f = T_f(1 + e_a) + \frac{t_{proc}(e_a - e_b)}{2}$$

The error is $\hat{T}_f - T_f$, so:

$$Error = T_f \times e_a + \frac{t_{proc}(e_a - e_b)}{2}$$

The time of flight is about 33 nanoseconds, however $t_{proc}$ is around $3,000,000$ns.
So we can ignore $T_f \times e_a$ cause $T_f \ll t_{proc}$, and we obtain :

$$Error \approx \frac{t_{proc}(e_a - e_b)}{2} \tag{IV.2}$$

### DS-TWR Error calculation :

Let's set:

- True time of flight : $T_f$
- True times of process : $t_{pa}$ and $t_{pb}$
- Clock drift errors : $e_a$ and $e_b$

The measured times are:

$$R_a = (2T_f + t_{pb})(1 + e_a), \quad D_a = t_{pa}(1 + e_a)$$

$$R_b = (2T_f + t_{pa})(1 + e_b), \quad D_b = t_{pb}(1 + e_b)$$

The calculated time of flight is:

$$\hat{T}_f = \frac{R_a R_b - D_a D_b}{R_a + R_b + D_a + D_b}$$

By replacing the measured times, we obtain :

$$\hat{T}_f = \frac{(1 + e_a)(1 + e_b) \times 2T_f(2T_f + t_{pa} + t_{pb})}{(2 + e_a + e_b)(2T_f + t_{pa} + t_{pb})}$$

By simplifying, we obtain :

$$\hat{T}_f = \frac{2T_f(1 + e_a)(1 + e_b)}{2 + e_a + e_b} \approx T_f \left(1 + \frac{e_a + e_b}{2}\right)$$

The error is $\hat{T}_f - T_f$, so:

$$Error \approx T_f \times \frac{e_a + e_b}{2} \tag{IV.3}$$

# References


[1] S. Gezici, H. Kobayashi, H. V. Poor, and Z. Sahinoglu, "Localization via Ultra-Wideband Radios," en,

[2] *(PDF) THE ALOHA SYSTEM: Another alternative for computer communications*, en. DOI: 10.1145/1478462.1478502 Accessed: Oct. 13, 2026. [Online]. Available: https://www.researchgate.net/publication/234790393_THE_ALOHA_SYSTEM_another_alternative_for_computer_communications

[3] "(PDF) ALOHA packet system with and without slots and capture," en, *ResearchGate*, Aug. 2026. DOI: 10.1145/1024916.1024920 Accessed: Oct. 13, 2026. [Online]. Available: https://www.researchgate.net/publication/234802142_ALOHA_packet_system_with_and_without_slots_and_capture

[4] R. E. Kalman, "A New Approach to Linear Filtering and Prediction Problems," en, *Journal of Basic Engineering*, vol. 82, no. 1, pp. 35–45, Mar. 1960, ISSN: 0021-9223. DOI: 10.1115/1.3662552 Accessed: Oct. 5, 2026. [Online]. Available: https://asmedigitalcollection.asme.org/fluidsengineering/article/82/1/35/397706/A-New-Approach-to-Linear-Filtering-and-Prediction

[5] G. Welch and G. Bishop, "An Introduction to the Kalman Filter," University of North Carolina at Chapel Hill, Tech. Rep. TR 95-041, 2006.

[6] M. Boutayeb, H. Rafaralahy, and M. Darouach, "Convergence analysis of the extended Kalman filter used as an observer for nonlinear deterministic discrete-time systems," *IEEE Transactions on Automatic Control*, vol. 42, no. 4, pp. 581–586, Apr. 1997, ISSN: 00189286. DOI: 10.1109/9.566674 Accessed: Oct. 6, 2026. [Online]. Available: http://ieeexplore.ieee.org/document/566674/

[7] S. J. Julier and J. K. Uhlmann, "New extension of the Kalman filter to nonlinear systems," I. Kadar, Ed., Orlando, FL, USA, Jul. 1997, p. 182. DOI: 10.1117/12.280797 Accessed: Oct. 13, 2026. [Online]. Available: http://proceedings.spiedigitallibrary.org/proceeding.aspx?doi=10.1117/12.280797

[8] M. Arulampalam, S. Maskell, N. Gordon, and T. Clapp, "A tutorial on particle filters for online nonlinear/non-Gaussian Bayesian tracking," *IEEE Transactions on Signal Processing*, vol. 50, no. 2, pp. 174–188, Feb. 2002, ISSN: 1053587X. DOI: 10.1109/78.978374 Accessed: Oct. 13, 2026. [Online]. Available: http://ieeexplore.ieee.org/document/978374/

[9] Y. Cao, M. Li, I. Svogor, S. Wei, and G. Beltrame, "Dynamic Range-Only Localization for Multi-Robot Systems," *IEEE Access*, vol. 6, pp. 46 527–46 537, 2018, ISSN: 2169-3536. DOI: 10.1109/ACCESS.2018.2866259 Accessed: Oct. 6, 2026. [Online]. Available: https://ieeexplore.ieee.org/document/8443328/

[10] K. J. Åström and R. M. Murray, *Feedback systems: an introduction for scientists and engineers*, Second edition. Princeton: Princeton University Press, 2021, ISBN: 978-0-691-19398-4.

[11] S. Van Der Helm, M. Coppola, K. N. McGuire, and G. C. H. E. De Croon, "On-board range-based relative localization for micro air vehicles in indoor leader–follower flight," en, *Autonomous Robots*, vol. 44, no. 3-4, pp. 415–441, Mar. 2020, ISSN: 0929-5593, 1573-7527. DOI: 10.1007/s10514-019-09843-6 Accessed: Oct. 13, 2026. [Online]. Available: http://link.springer.com/10.1007/s10514-019-09843-6

[12] S. Li, F. Shan, J. Liu, M. Coppola, C. De Wagter, and G. C. H. E. De Croon, "Onboard Ranging-Based Relative Localization and Stability for Lightweight Aerial Swarms," *IEEE Robotics and Automation Letters*, vol. 10, no. 10, pp. 10 066–10 073, Oct. 2025, ISSN: 2377-3766, 2377-3774. DOI: 10.1109/LRA.2025.3597039 Accessed: Oct. 9, 2026. [Online]. Available: https://ieeexplore.ieee.org/document/11119828/

[13] Y.-C. Chen, A. I.-C. Lai, and R.-B. Wu, "UWB-Assisted High-Precision Positioning in a UTM Prototype," in *2020 IEEE Topical Conference on Wireless Sensors and Sensor Networks (WiSNeT)*, San Antonio, TX, USA: IEEE, Jan. 2020, pp. 42–45, ISBN: 978-1-7281-3355-3. DOI: 10.1109/WiSNeT46826.2020.9037604 Accessed: Oct. 13, 2026. [Online]. Available: https://ieeexplore.ieee.org/document/9037604/

[14] J. Yang and S. Lee, "Ultrawideband coupled relative positioning algorithm applicable to flight controller for multidrone collaboration," en, *ETRI Journal*, vol. 45, no. 5, pp. 758–767, Oct. 2023, ISSN: 1225-6463, 2233-7326. DOI: 10.4218/etrij.2023-0128 Accessed: Oct. 12, 2026. [Online]. Available: https://onlinelibrary.wiley.com/doi/10.4218/etrij.2023-0128

[15] *IEEE Standards Association*, en. Accessed: Nov. 13, 2026. [Online]. Available: https://standards.ieee.org/ieee/802.15.4/7029/

[16] D. Eridani, A. F. Rochim, and F. N. Cesara, "Comparative Performance Study of ESP-NOW, Wi-Fi, Bluetooth Protocols based on Range, Transmission Speed, Latency, Energy Usage and Barrier Resistance: 2021 International Seminar on Application for Technology of Information and Communication, iSemantic 2021," *Proceedings - 2021 International Seminar on Application for Technology of Information and Communication*, Proceedings - 2021 International Seminar on Application for Technology of Information and Communication: IT Opportunities and Creativities for Digital Innovation and Communication within Global Pandemic, iSemantic 2021, pp. 322–328, Sep. 2021. DOI: 10.1109/iSemantic52711.2021.9573246 Accessed: Nov. 13, 2026. [Online]. Available: https://www.scopus.com/pages/publications/85118937413

[17] *ESP-NOW User Guide*. Accessed: Nov. 13, 2026. [Online]. Available: https://documentation.espressif.com/esp-now/master/User_Guide.md